

Rocket Nozzle CFD Simulation

Three-case study: hot gas · premixed combustion · separate injection

OpenFOAM v2012 · Axisymmetric wedge · 14 400 cells

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1. Overview

Three OpenFOAM simulations of a De Laval (converging-diverging) rocket nozzle are presented on an identical extended 5-block axisymmetric mesh. The domain extends from the chamber head ($x = -0.30\text{ m}$) through the nozzle throat and diverging section to a downstream plume region ($x = 1.50\text{ m}$), allowing the supersonic jet and its interaction with the ambient far-field to be captured.

Case	Solver	Description
Hot gas (frozen)	rhoCentralFoam	Single-species perfect gas ($M=22\text{ g/mol}$, $\gamma=1.2$). Density-based explicit solver.
Premixed combustion	rhoReactingFoam	$\text{C}_2\text{H}_2/\text{O}_2$ pre-mixed at inlet. Single-step Arrhenius. JANAF thermodynamics.
Separate injection	rhoReactingFoam	Fuel and LOX through dedicated inlets; mixing and reaction downstream.

2. Mesh

The mesh is a 5-degree wedge (one cell in the azimuthal direction) generated with **blockMesh**. Five blocks cover the chamber, converging section, diverging section, plume core ($x = 0.50\text{--}1.50\text{ m}$, $r = 0\text{--}0.14\text{ m}$), and plume ambient ($x = 0.50\text{--}1.50\text{ m}$, $r = 0.14\text{--}0.50\text{ m}$).

Parameter	Value
Chamber radius	0.150 m
Throat radius	0.050 m
Exit radius	0.1414 m
Throat position	$x = 0.150\text{ m}$
Domain x range	$-0.30\text{ m} \rightarrow 1.50\text{ m}$
Domain r range	$0 \rightarrow 0.50\text{ m}$
Total cells	14 400

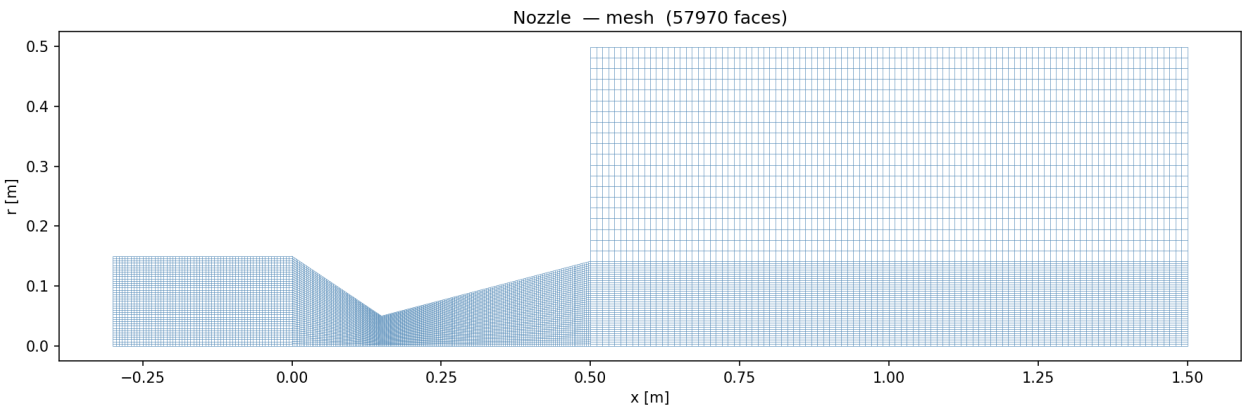


Fig. 1 — Axisymmetric wedge mesh (hot-gas case, x - r plane).

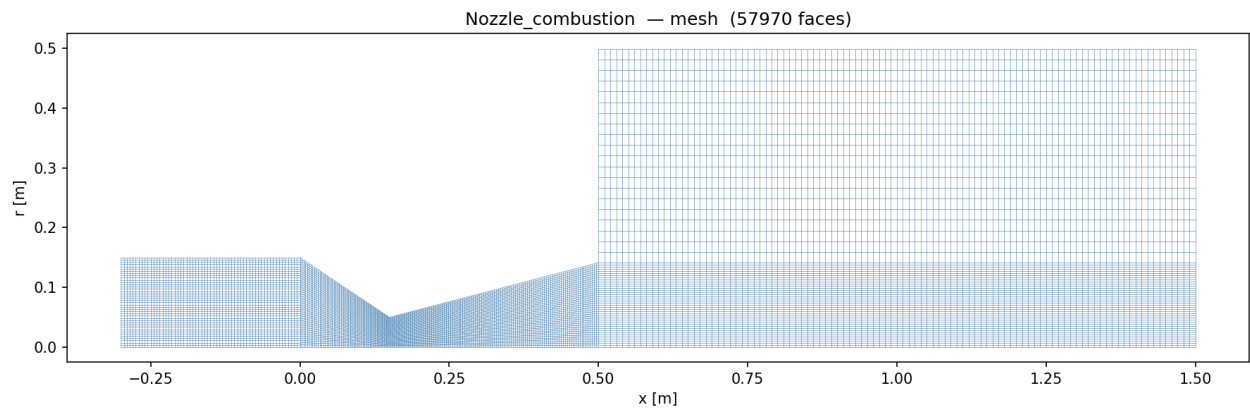


Fig. 2 — Same mesh used for combustion cases.

3. Case 1 — Hot Gas (rhoCentralFoam)

Combustion products are approximated as a single-species perfect gas representative of an LOX/kerosene flame ($M = 22 \text{ g/mol}$, $\gamma = 1.2$, $C_p = 2\,267 \text{ J/kg K}$). The density-based explicit solver **rhoCentralFoam** (Kurganov–Tadmor flux, vanLeer MUSCL, CFL = 0.1) captures the transonic throat, supersonic expansion fan, and plume development.

Property	Value	Notes
Molecular weight	22 g/mol	CO ₂ /H ₂ O/CO/H ₂ mixture
γ	1.2	Triatomic-dominated
Chamber pressure p_0	3 MPa	Total pressure at inlet
Chamber temperature T_0	3 000 K	Approximate adiabatic flame T
Back pressure	101 325 Pa	Sea level — over-expanded nozzle
Exit Mach (design)	3.20	Isentropic 1-D theory

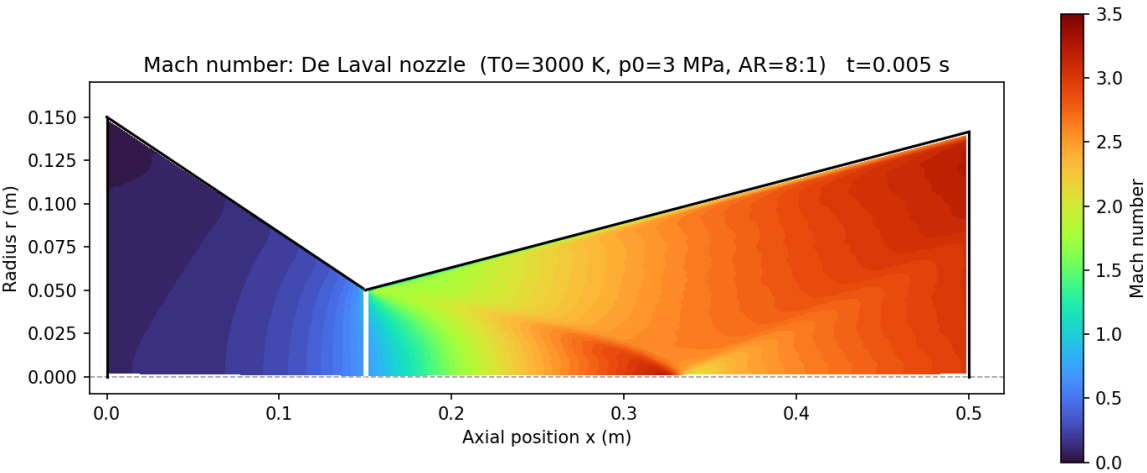


Fig. 3 — Mach number contour (nozzle-only domain, validated design point). Exit Mach ≈ 3.22 matches isentropic theory (3.20) to within 1%.

3.2 Extended plume domain — early transient ($t = 0.0005 \text{ s}$)

The hot-gas case on the extended 5-block plume domain is shown at $t = 0.5 \text{ ms}$. The inlet shock is propagating into the initially-atmospheric plume region. Long-time steady state is represented by the validated nozzle-only result above.

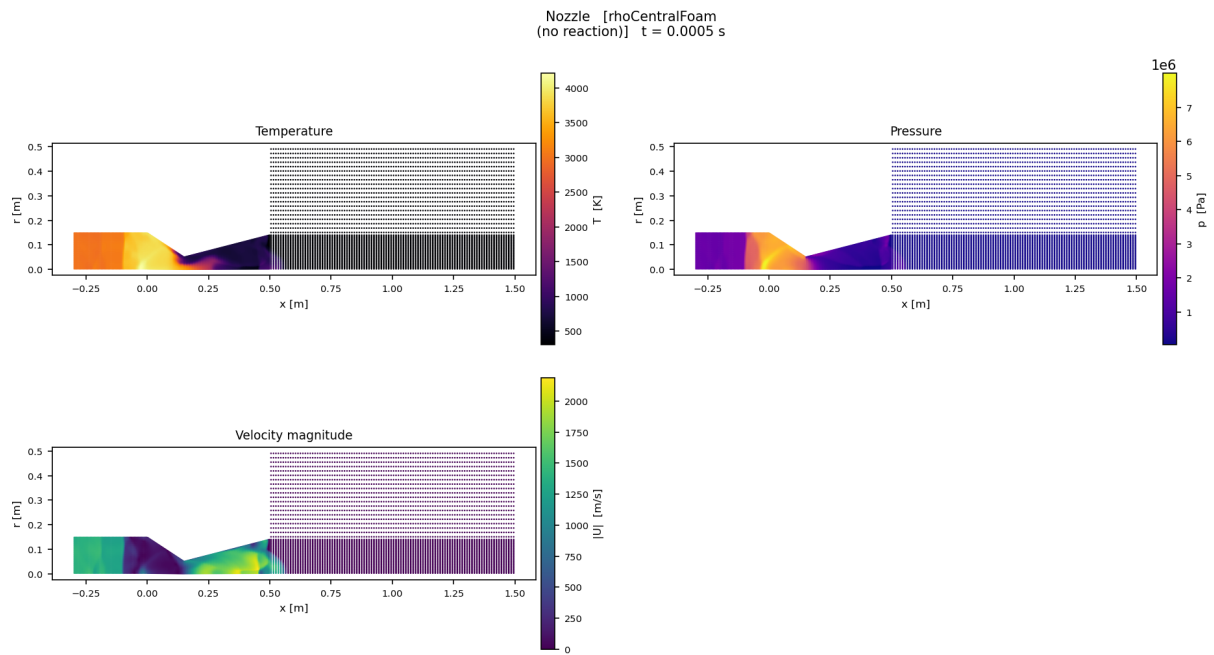


Fig. 4 — T, p, |U| at t = 0.5 ms (extended plume domain, startup transient).

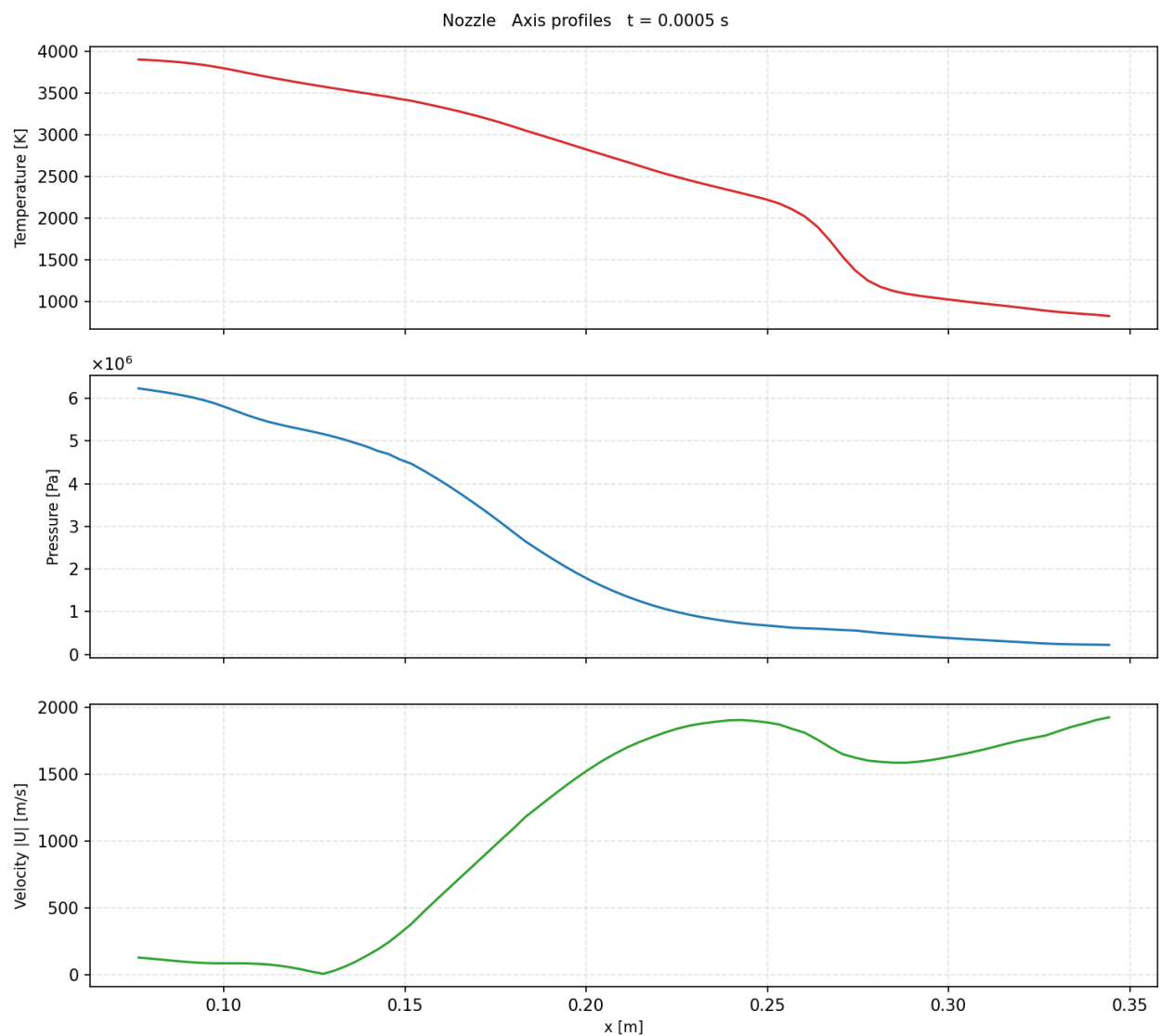
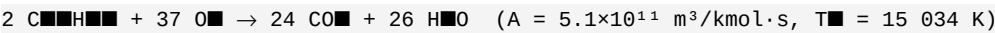


Fig. 5 — Axis profiles of T, p, |U| at t = 0.5 ms.

4. Case 2 — Premixed Combustion (rhoReactingFoam)

Pre-mixed $\text{C}_{12}\text{H}_{26}/\text{O}_2$ (kerosene surrogate; $Y_{\text{fuel}} = 0.20$, $Y_{\text{O}_2} = 0.80$) enters at $T = 500 \text{ K}$, $p = 3 \text{ MPa}$. A single-step global Arrhenius reaction releases heat as the reactants mix and ignite. Species thermodynamics use NASA-7 JANAF polynomials; viscosity follows Sutherland's law. Simulation time: $0 \rightarrow 0.003 \text{ s}$.

Reaction:



4.1 Flow fields — $t = 0.003 \text{ s}$

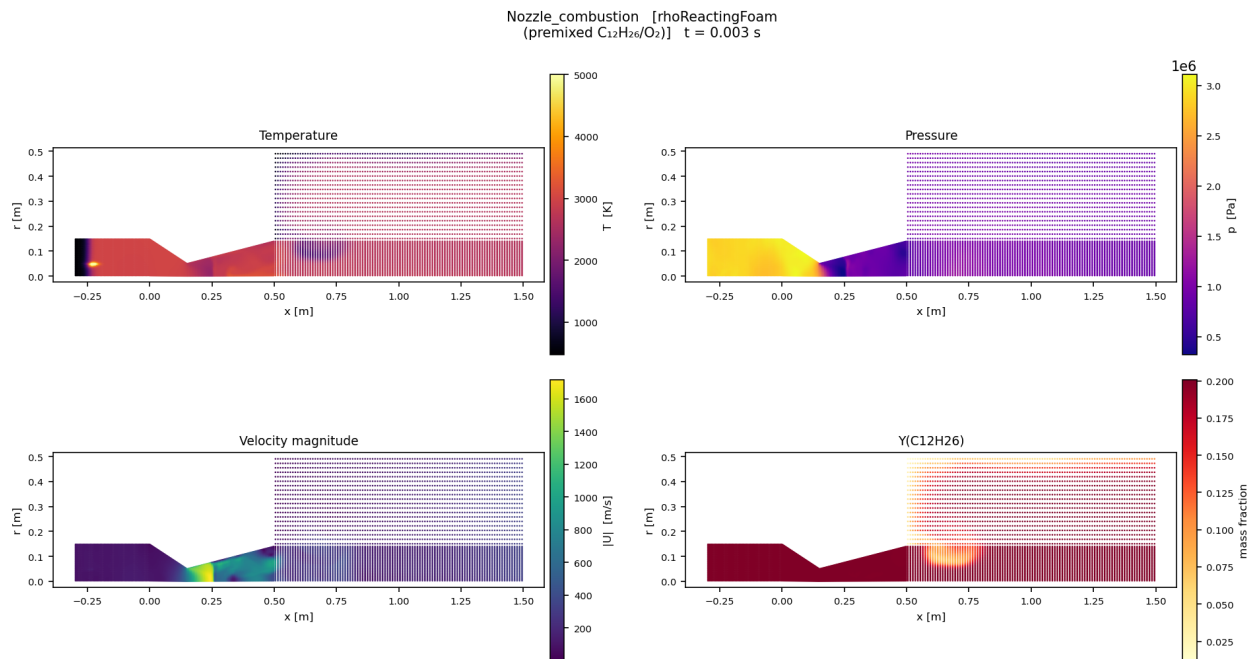


Fig. 4 — Temperature, pressure, velocity magnitude, and $\text{C}_{12}\text{H}_{26}$ mass fraction at $t = 0.003 \text{ s}$ (premixed case).

4.2 Axis profiles

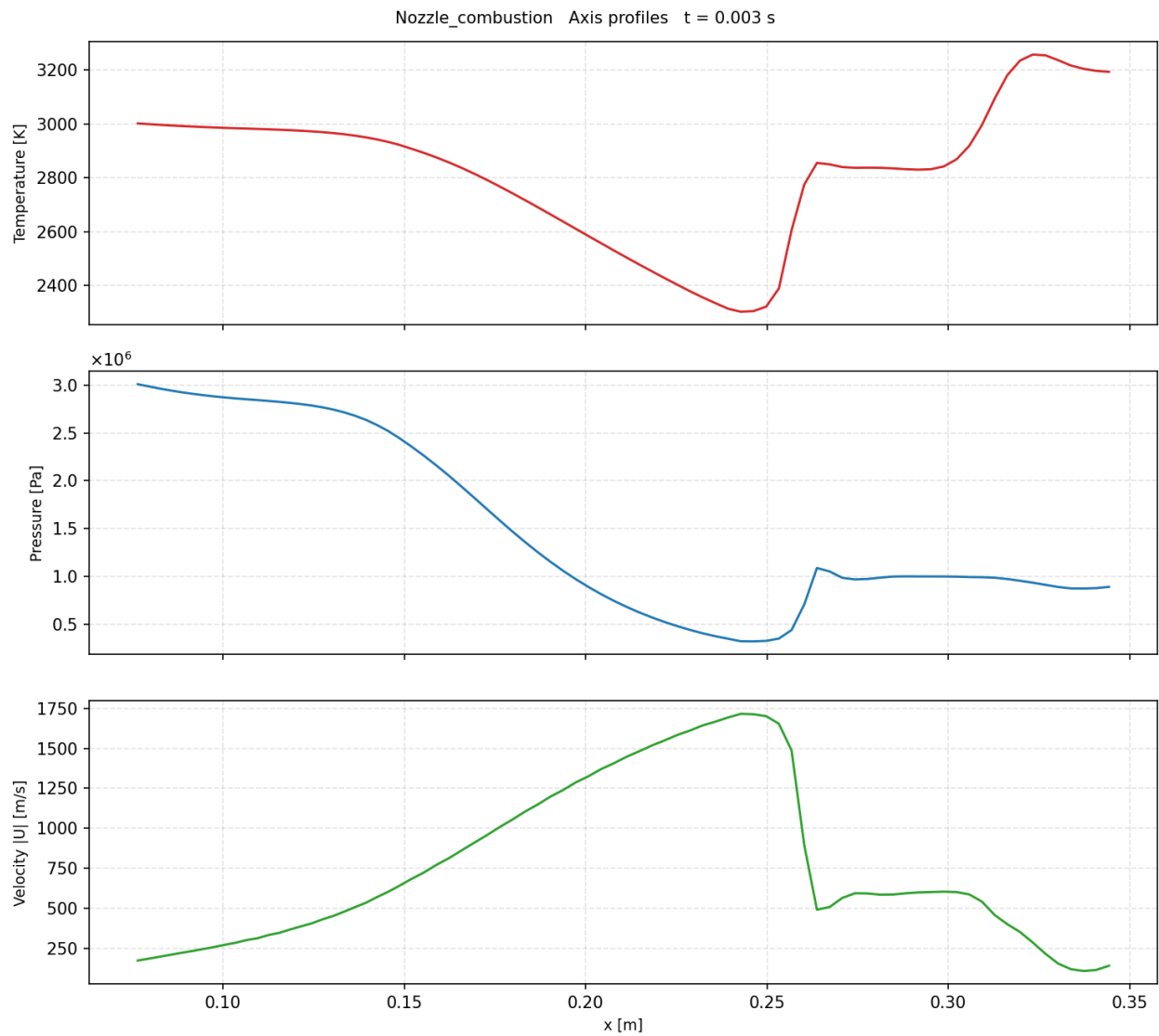


Fig. 5 — Temperature, pressure, and velocity magnitude along the nozzle axis (premixed case).

4.3 Species distributions

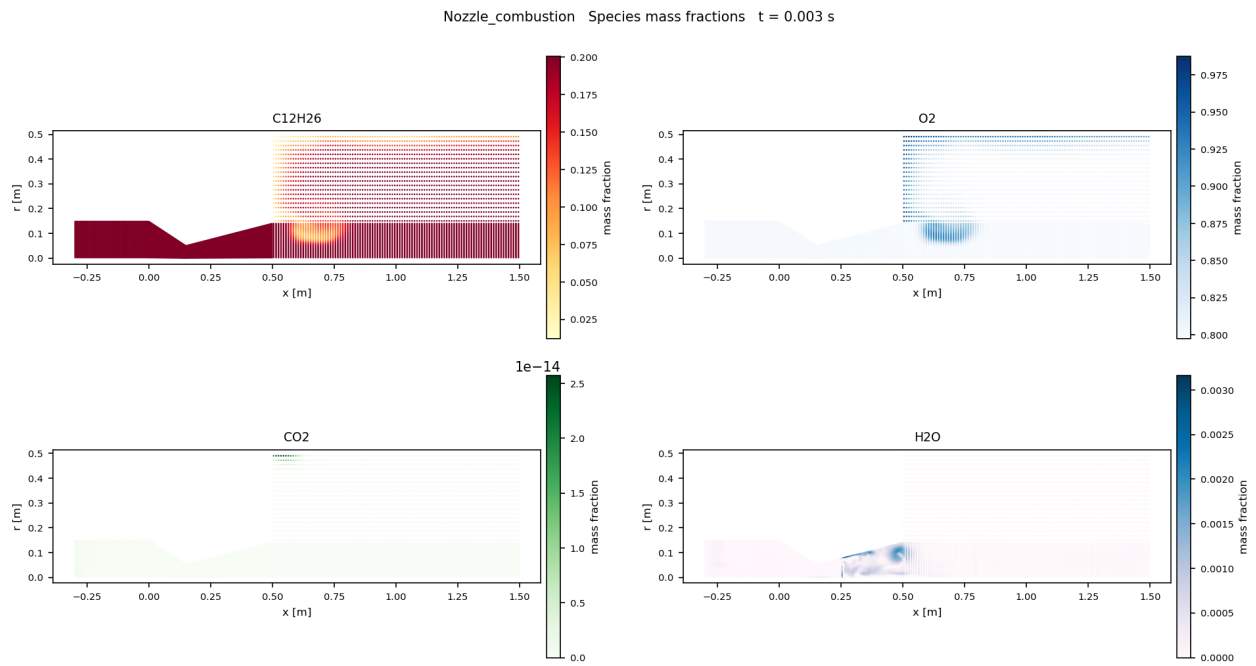


Fig. 6 — Mass fraction fields (C $\blacksquare$ H $\blacksquare$ H $\blacksquare$, O $\blacksquare$, CO $\blacksquare$, H $\blacksquare$ O) at $t = 0.003$ s.

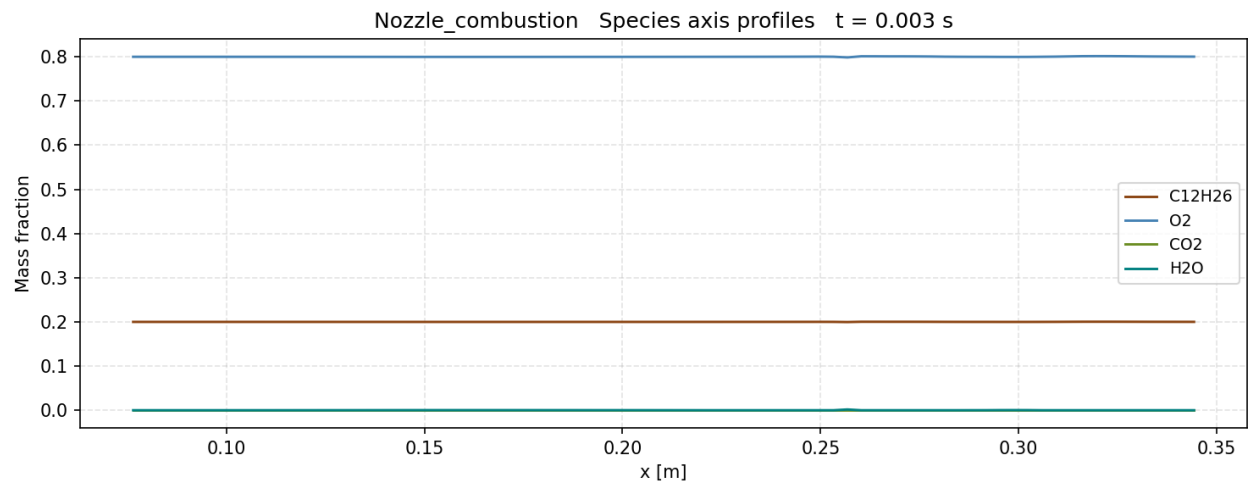


Fig. 7 — Species mass fractions along the nozzle axis.

5. Case 3 — Separate Injection (rhoReactingFoam)

Kerosene (fuel_inlet) and liquid oxygen (lox_inlet) are injected through separate concentric annular inlets at the chamber head. The chamber is pre-filled with combustion equilibrium products ($\text{CO}_2 = 0.693$, $\text{H}_2\text{O} = 0.307$ by mass) at $T = 3\,000\text{ K}$ and $p = 3\text{ MPa}$. Reaction occurs downstream where mixing brings reactants to the ignition threshold. Simulation time: $0 \rightarrow 0.005\text{ s}$.

Parameter	Value
Fuel inlet species	$\text{C}_{12}\text{H}_{26}$ (kerosene surrogate)
Oxidiser inlet species	O_2
Inlet total pressure	3 MPa (both)
Design O/F ratio	≈ 2.7 (stoichiometric $\text{C}_{12}\text{H}_{26}/\text{O}_2$)
Initial chamber fill	$\text{CO}_2 + \text{H}_2\text{O}$ (equilibrium products)

5.1 Flow fields — $t = 0.005\text{ s}$

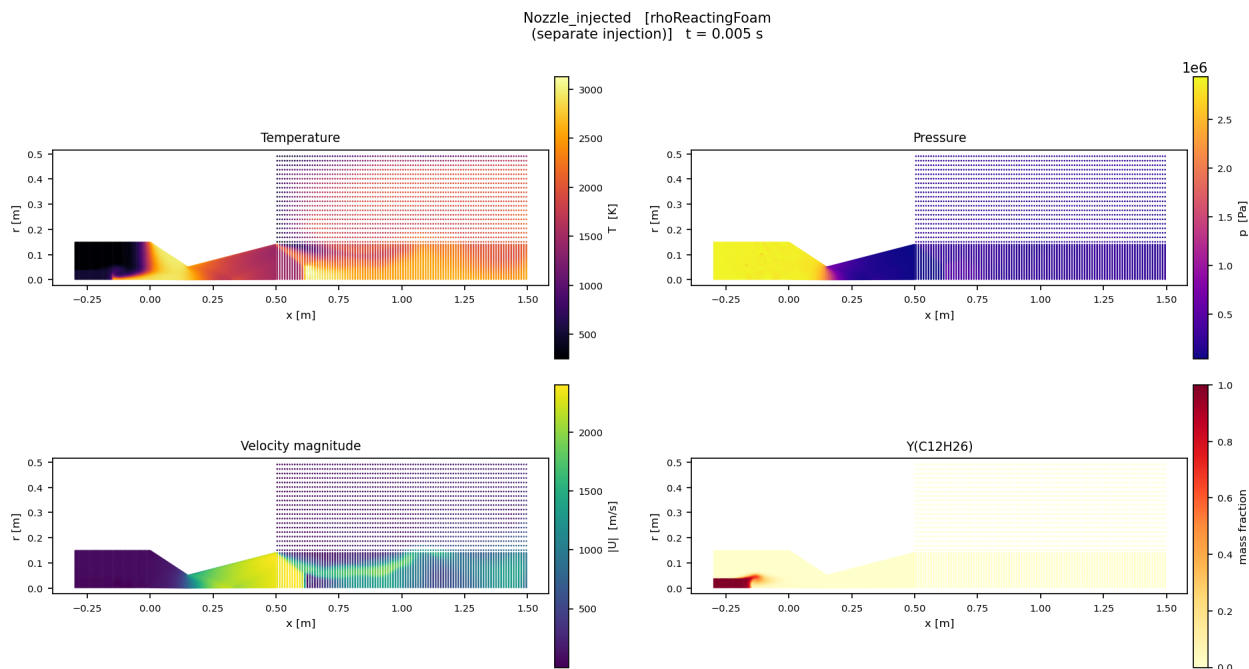


Fig. 8 — Temperature, pressure, velocity magnitude, and $\text{C}_{12}\text{H}_{26}$ mass fraction at $t = 0.005\text{ s}$ (separate injection case).

5.2 Axis profiles

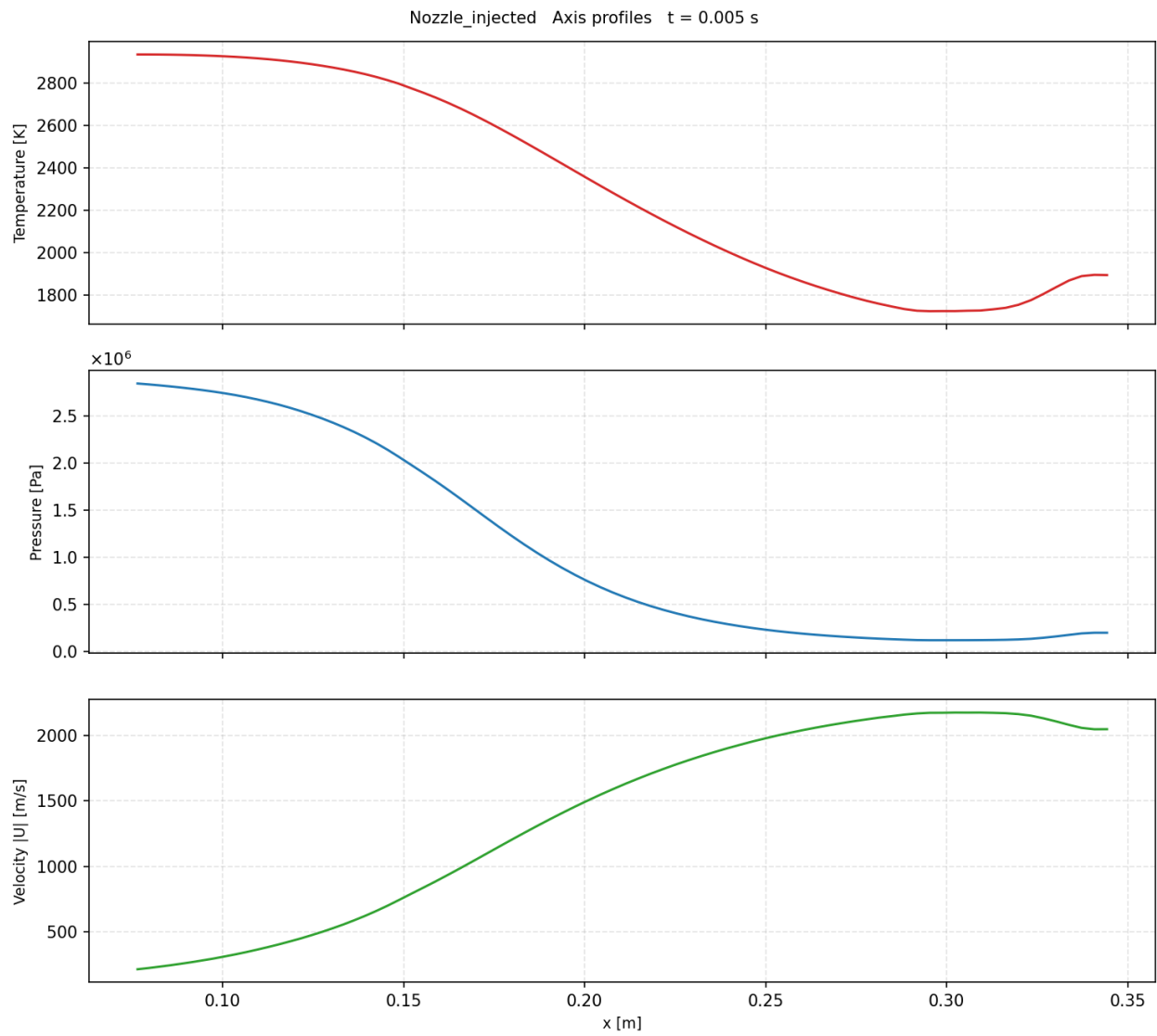


Fig. 9 — Temperature, pressure, and velocity magnitude along the nozzle axis (separate injection case).

5.3 Species distributions

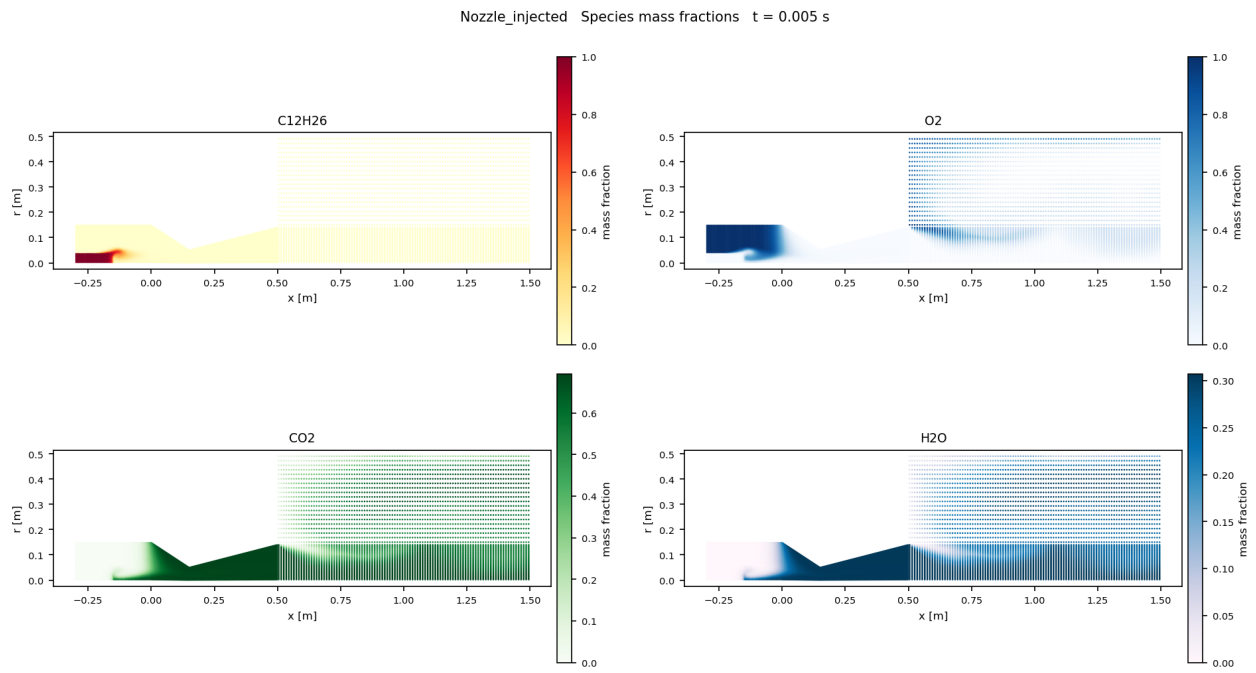


Fig. 10 — Mass fraction fields (C₁₂H₂₆, O₂, CO₂, H₂O) at $t = 0.005$ s.

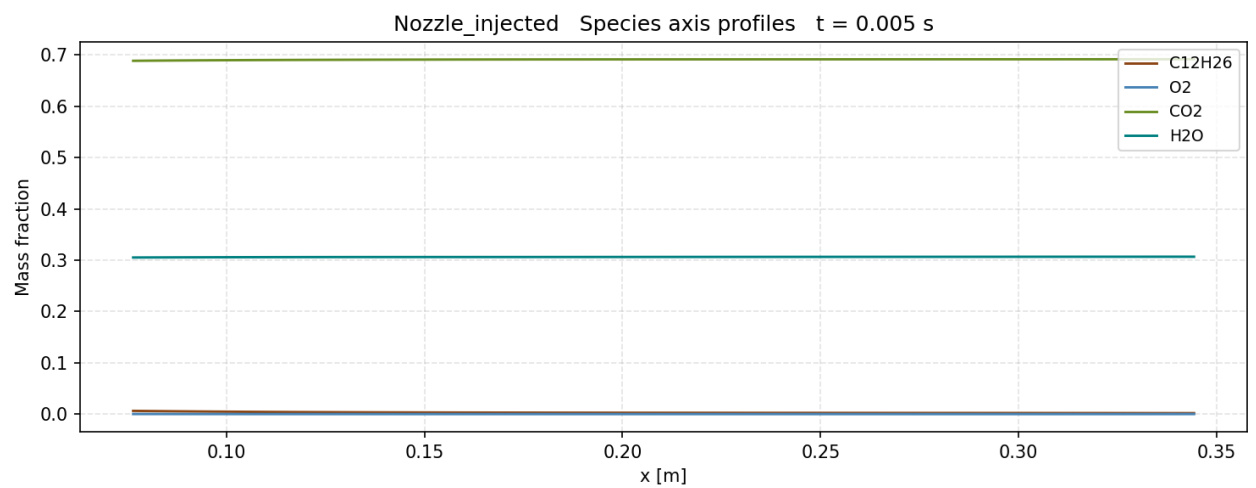


Fig. 11 — Species axis profiles showing fuel/oxidiser mixing and product formation.

6. Implementation Notes

- **Thermodynamic consistency (rhoCentralFoam):** The initial density must satisfy $\rho = p / (R_{\text{spec}} \cdot T)$ uniformly. Mismatching the far-field p against the interior pressure causes Newton iteration in `hePsiThermo::correct()` to diverge to $T \rightarrow 10^{12}$ K. Outer boundary: `zeroGradient` on both p and ρ ensures consistency at startup.
- **foamChemistryReader elements block:** `readSpeciesComposition()` looks for an 'elements {}' sub-dictionary directly inside each species' top-level block in `constant/thermo` — NOT inside the nested 'specie {}' block. A top-level 'elements (C H O);' list also triggers a `HashPtrTable` parsing error.
- **Species mass fraction initialisation:** All Y_i must sum to 1 at every patch face at $t = 0$ (including inletOutlet 'value' fields at the outlet). A zero-sum causes division-by-zero in `multiComponentMixture::patchFaceVolMixture()`.
- **Far-field pressure BC:** `waveTransmissive` at the outer_boundary amplifies the startup pressure wave from the high-pressure interior into the atmospheric far-field. `zeroGradient` for p at that boundary is more stable during the initial transient.
- **C12H26 JANAF coefficients:** `Tcommon` is set to 300 K with identical low/high coefficient arrays to avoid discontinuity artefacts; $T_{\text{low}} = 200$ K, $T_{\text{high}} = 5\,000$ K. The `JANAF limit()` function clips T to $[T_{\text{low}}, T_{\text{high}}]$ — out-of-range warnings are expected in regions exceeding 5 000 K and do not indicate solver failure.

7. Limitations

- Conical nozzle — a bell or Rao-optimised profile reduces divergence loss.
- Single-step global Arrhenius — no intermediate species (CO, OH, H) or dissociation.
- Adiabatic walls — no regenerative cooling or conjugate heat transfer.
- Laminar — turbulent mixing of the fuel/oxidiser shear layer not modelled (Case 3).
- 2-D axisymmetric — no azimuthal instabilities, swirl, or injector pattern effects.
- Short simulation times — steady-state plume not yet established in Cases 2 and 3.